Table 1: Average acceptance length τ, decoding speed (tokens/s), and speedups across different models and settings on English long context datasets. (VLLM)

batchsize 100 , output 512

Model	Setting	GovReport				QMSum				Multi-News				LCC				RepoBench-P				Overall			
		Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑
Q3-4B	Vanilla	1,000	1,000x	1,000x	310,370	1,000	1,000x	1,000x	338,070	1,000	1,000x	1,000x	303,860	1,000	1,000x	1,000x	336,360	1,000	1,000x	1,000x	312,99 / 311,82	1,000	1,000x	1,000x	321,20 / 320,40
	EAGLE3	2,250	1,102x	1,139x	352,610	2,050	1,038x	1,060x	357,530	2,020	1,018x	1,024x	310,950	2,080	1,040x	1,062x	356,130	2,040	1,023x	1,030x	321,130	2,088	1,045x	1,064x	340,800
	DFlash	2,120	1,398x	1,594x	494,560	2,520	1,673x	1,830x	618,430	1,820	1,231x	1,340x	407,170	2,820	1,811x	2,023x	680,270	2,420	1,493x	1,728x	540,830	2,334	1,495x	1,689x	542,500
	Domino	1,080	0,701x	0,641x	198,530	1,090	0,656x	0,633x	213,450	1,060	0,706x	0,637x	193,500	1,050	0,639x	0,613x	205,510	1,050	0,680x	0,625x	194,780	1,066	0,676x	0,629x	201,600
	DSpark	**3,060**	**1,724x**	**2,216x**	**687,760**	**3,070**	**2,012x**	**2,300x**	**777,600**	**2,560**	**1,525x**	**1,868x**	**567,540**	**3,290**	**2,074x**	**2,396x**	**805,800**	**3,110**	**1,764x**	**2,228x**	**697,350**	**3,027**	**1,799x**	**2,203x**	**707,500**
	Ours																								

Table 2: Average acceptance length τ, decoding speed (tokens/s), and speedups across different models and settings on VietNamese summarization datasets. (VLLM)

Model	Setting	VietNews				WikiLingua				VLMs				VLSP				Overall			
		Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑
Q3-4B	Vanilla	1,000	1,000x	1,000x	349,53	1,000	1,000×	1,000×	350,39	1,000	1,000×	1,000×	335,23	1,000	1,000×	1,000×	338,07	1,000	1,000×	1,000×	343,17
	EAGLE3	1,346	0,680x	0,682x	238,27	1,252	0,622×	0,631×	221,19	1,320	0,682×	0,672×	225,28	1,312	0,681×	0,671×	226,76	1,307	0,667×	0,664×	227,70
	DFlash	2,178	1,564x	1,586x	554,32	1,882	1,324×	1,374×	481,46	1,918	1,362×	1,408×	471,92	1,925	1,379×	1,402×	473,93	1,976	1,403×	1,437×	493,18
	Domino	1,111	0,632x	0,634x	221,43	1,130	0,629×	0,641×	224,77	1,092	0,644×	0,632×	211,95	1,094	0,642×	0,632×	213,72	1,107	0,637×	0,635×	217,86
	DSpark	**2,305**	**1,696x**	**1,715x**	**599,52**	**2,023**	**1,447×**	**1,496×**	**524,11**	**2,176**	**1,553×**	**1,607×**	**538,79**	**2,145**	**1,520×**	**1,584×**	**535,62**	**2,162**	**1,551×**	**1,597×**	**548,09**
	Ours																				

Ở 2 bảng trên thì các phương pháp sử dụng verify theo tree đều nhận kết quả chưa OK có thể do cơ chế batching của VLLM thực hiện khác với transformer

Table 3: Average acceptance length τ, decoding speed (tokens/s), and speedups across different models and settings on VietNamese summarization datasets. (FA4)

8192 input , 512 output

Model	Setting	VietNews				WikiLingua				VLMs				VLSP				Overall			
		Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑
Q3-4B	Vanilla	1,000	1,000x	1,000x	36,03	1,000	1,000×	1,000×	35,52	1,000	1,000×	1,000×	35,55	1,000	1,000×	1,000×	35,47	1,000	1,000×	1,000×	35,64
	EAGLE3	1,41	1,103x	1,103x	40,23	1,32	1,071×	1,072×	38,00	1,38	1,100×	1,101×	39,46	1,37	1,049×	1,050×	38,73	1,37	1,080×	1,081×	39,11
	DFlash	2,26	1,785x	1,796x	**66,69**	1,93	**1,537×**	**1,545×**	**55,78**	2,02	**1,595×**	**1,611×**	**58,75**	1,99	**1,519×**	**1,528×**	**56,74**	2,05	**1,602×**	**1,613×**	**59,49**
	Domino	2,23	1,536x	1,542x	57,57	1,91	1,342×	1,346×	48,50	2,08	1,396×	1,406×	51,61	2,01	1,389×	1,395×	51,08	2,06	1,412×	1,418×	52,19
	DSpark	**2,30**	**1,801x**	**1,812x**	66,63	**2,02**	1,496×	1,503×	54,27	**2,18**	1,474×	1,486×	54,39	**2,14**	1,489×	1,498×	54,78	**2,16**	1,553×	1,563×	57,52
	Ours																				

Table 4: Average acceptance length τ, decoding speed (tokens/s), and speedups across different models and settings on English long context datasets. (FA4)

8192 input , 512 output

Model	Setting	GovReport				QMSum				Multi-News				LCC				RepoBench-P				Overall			
		Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑
Q3-4B	Vanilla	1,000	1,000x	1,000x	33,830	1,000	1,000x	1,000x	33,850	1,000	1,000x	1,000x	35,310	1,000	1,000x	1,000x	34,260	1,000	1,000x	1,000x	34,890	1,000	1,000x	1,000x	34,430
	EAGLE3	2,170	1,712x	1,725x	58,900	2,340	1,791x	1,857x	60,870	2,550	2,049x	2,061x	72,710	2,800	2,165x	2,183x	75,440	2,520	2,018x	2,035x	71,260	2,480	1,947x	1,972x	67,830
	DFlash	2,190	1,695x	1,707x	57,880	1,950	1,448x	1,478x	49,430	2,460	1,966x	1,976x	70,260	4,370	3,257x	3,308x	**121,260**	3,740	2,735x	2,773x	100,920	2,940	2,220x	2,248x	79,950
	Domino	2,610	1,759x	1,773x	60,230	2,430	1,536x	1,574x	53,470	2,850	2,003x	2,013x	71,350	**4,680**	2,890x	2,928x	113,300	**4,290**	2,799x	2,840x	102,170	3,370	2,197x	2,226x	80,100
	DSpark	**2,880**	**2,160x**	**2,187x**	**74,110**	**2,670**	**1,950x**	**2,038x**	**66,740**	**3,030**	**2,430x**	**2,448x**	**86,360**	4,350	**3,284x**	**3,336x**	115,830	4,150	**3,094x**	**3,147x**	**110,710**	**3,420**	**2,584x**	**2,631x**	**90,750**
	Ours																								

Table 5: Average acceptance length τ, decoding speed (tokens/s), and speedups across different models and settings on various length bins English long context data. (VLLM)

Model	Setting	Bin 1 (0-2k)				Bin 2 (2-4k)				Bin 3 (4-8k)				Bin 4 (8-12k)				Bin 5 (12-16k)				Overall			
		Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑
Q3-4B	Vanilla	1,000	1,000x	1,000x	342,279	1,000	1,000x	1,000x	335,001	1,000	1,000x	1,000x	321,773	1,000	1,000x	1,000x	311,535	1,000	1,000x	1,000x	303,435	1,000	1,000x	1,000x	322,805
	EAGLE3	2,183	1,113x	1,115x	381,658	2,158	1,108x	1,111x	372,048	2,117	1,085x	1,089x	350,322	2,127	1,088x	1,095x	341,221	2,099	1,080x	1,090x	330,790	2,137	1,095x	1,100x	355,208
	DFlash	3,450	2,313x	2,351x	804,861	3,629	2,340x	2,398x	803,482	2,674	1,776x	1,832x	589,403	2,531	1,605x	1,681x	523,768	2,258	1,478x	1,565x	474,758	2,908	1,902x	1,966x	639,254
	Domino	1,059	0,620x	0,617x	211,154	1,051	0,622x	0,618x	206,929	1,060	0,635x	0,626x	201,347	1,055	0,644x	0,627x	195,358	1,063	0,665x	0,640x	194,300	1,058	0,637x	0,626x	201,818
	DSpark	**3,834**	**2,733x**	**2,794x**	**956,220**	**3,841**	**2,707x**	**2,796x**	**936,587**	**3,390**	**2,360x**	**2,493x**	**802,292**	**3,182**	**2,101x**	**2,291x**	**713,642**	**2,981**	**1,930x**	**2,164x**	**656,722**	**3,446**	**2,364x**	**2,508x**	**813,093**
	Ours																								

Table 6: Average acceptance length τ, decoding speed (tokens/s), and speedups across different models and settings on various length bins English long context data. (FA4)

Model	Setting	Bin 1 (0-2k)				Bin 2 (2-4k)				Bin 3 (4-8k)				Bin 4 (8-12k)				Bin 5 (12-16k)				Overall			
		Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑	Accept. Len. ↑	E2E Speedup ↑	Decode Speedup ↑	tok/s ↑
Q3-4B	Vanilla	1,000	1,000x	1,000x	32,476	1,000	1,000x	1,000x	32,159	1,000	1,000x	1,000x	31,891	1,000	1,000x	1,000x	31,334	1,000	1,000x	1,000x	31,546	1,000	1,000x	1,000x	31,881
	EAGLE3	2,760	2,224x	2,235x	73,027	2,660	2,082x	2,094x	68,477	2,500	1,974x	2,001x	64,137	1,860	1,426x	1,435x	46,543	1,330	0,992x	0,991x	31,596	2,220	1,587x	1,598x	56,756
	DFlash	3,490	2,331x	2,344x	87,616	3,600	2,528x	2,548x	93,832	2,700	1,922x	1,947x	68,157	2,560	1,699x	1,713x	59,680	2,230	1,411x	1,425x	48,645	2,910	1,889x	1,910x	71,586
	Domino	**3,900**	2,284x	2,295x	88,177	**3,900**	2,465x	2,484x	88,706	3,190	2,026x	2,056x	70,125	3,050	1,818x	1,842x	63,074	2,640	1,416x	1,431x	49,812	3,340	1,927x	1,949x	71,979
	DSpark	3,810	**2,868x**	**2,889x**	**98,657**	3,770	**2,874x**	**2,902x**	**97,323**	**3,300**	**2,368x**	**2,415x**	**80,746**	**3,240**	**2,249x**	**2,294x**	**74,703**	**2,980**	**1,849x**	**1,888x**	**62,483**	**3,420**	**2,375x**	**2,415x**	**82,782**
	Ours																								